Movie Rental
Data Warehouse
Full Pipeline Report

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Course: Data Architecture
Source OLTP: Sakila MySQL Database
Target DW: sakila_dw | Star Schema
Tools: XAMPP MySQL | Python 3 | mysql-connector-python

1. Introduction

This report documents the complete data warehouse design and implementation for a movie rental business using the Sakila OLTP MySQL database as the source system.

The Sakila schema is a 16 table normalised (3NF) operational database designed for daily transaction processing recording rentals, payments, customer registrations, and inventory management. While efficient for inserts and updates, it is not optimised for analytical queries such as revenue trends, film popularity, or store performance comparisons.

This project transforms Sakila into a star-schema data warehouse (sakila_dw) using a three-step pipeline:

Step	File	What it does
1	step1_create_dw_schema.sql	Creates sakila_dw database with 5 dimension tables and 2 fact tables
2	movie_rental_dw.ipynb	Python ETL: extracts from sakila, transforms data, loads sakila_dw and Runs 11 analytical business queries against the data warehouse

2. Business Questions

Eleven analytical business questions were identified from the Sakila schema to drive the data warehouse design:

#	Business Question	Business Value
BQ-1	Which films are rented most frequently?	Inventory planning and film procurement
BQ-2	Which films generate the highest revenue?	Pricing strategy and premium title selection
BQ-3	Which film categories are most popular?	Category-level marketing focus
BQ-4	Which stores generate the highest rentals and revenue?	Store performance benchmarking
BQ-5	Which customers rent the most and generate highest revenue?	Customer loyalty programmes
BQ-6	How does rental activity change by month and year?	Seasonal planning and budgeting
BQ-7	Which staff process the most rentals and payments?	Staff performance evaluation
BQ-8	Which countries and cities have the most active customers?	Geographic market expansion
BQ-9	What is the average rental duration per category?	Rental policy optimisation
BQ-10	Which films are returned late most often?	Late-fee revenue and customer satisfaction
BQ-11	How does quarterly revenue trend over time?	Financial reporting and forecasting

3. Dimensional Model Design

3.1 Schema Choice: Star Schema

A star schema was selected over a snowflake schema because it minimises joins at query time, making analytical queries faster and simpler. Geography chains (address ,city ,country) are fully denormalised into each dimension row.

3.2 Fact Tables

fact_rental Film Rental Transactions

Property	Value	Details
Grain	One row per rental	Each rental_id in sakila.rental = one row (16,044 rows loaded)
Dimensions	6 foreign keys	rental_date_key, return_date_key, customer_key, film_key, store_key, staff_key
Measures	4 metrics	rental_duration_days, expected_duration_days, is_late_return, days_late
Source Tables	OLTP	rental, inventory, film

fact_payment Payment Transactions

Property	Value	Details
Grain	One row per payment	Each payment_id in sakila.payment = one row (16,044 rows loaded)
Dimensions	5 foreign keys	payment_date_key, customer_key, staff_key, film_key, store_key
Measures	2 metrics	payment_amount, payment_count
Source Tables	OLTP	payment, rental, inventory

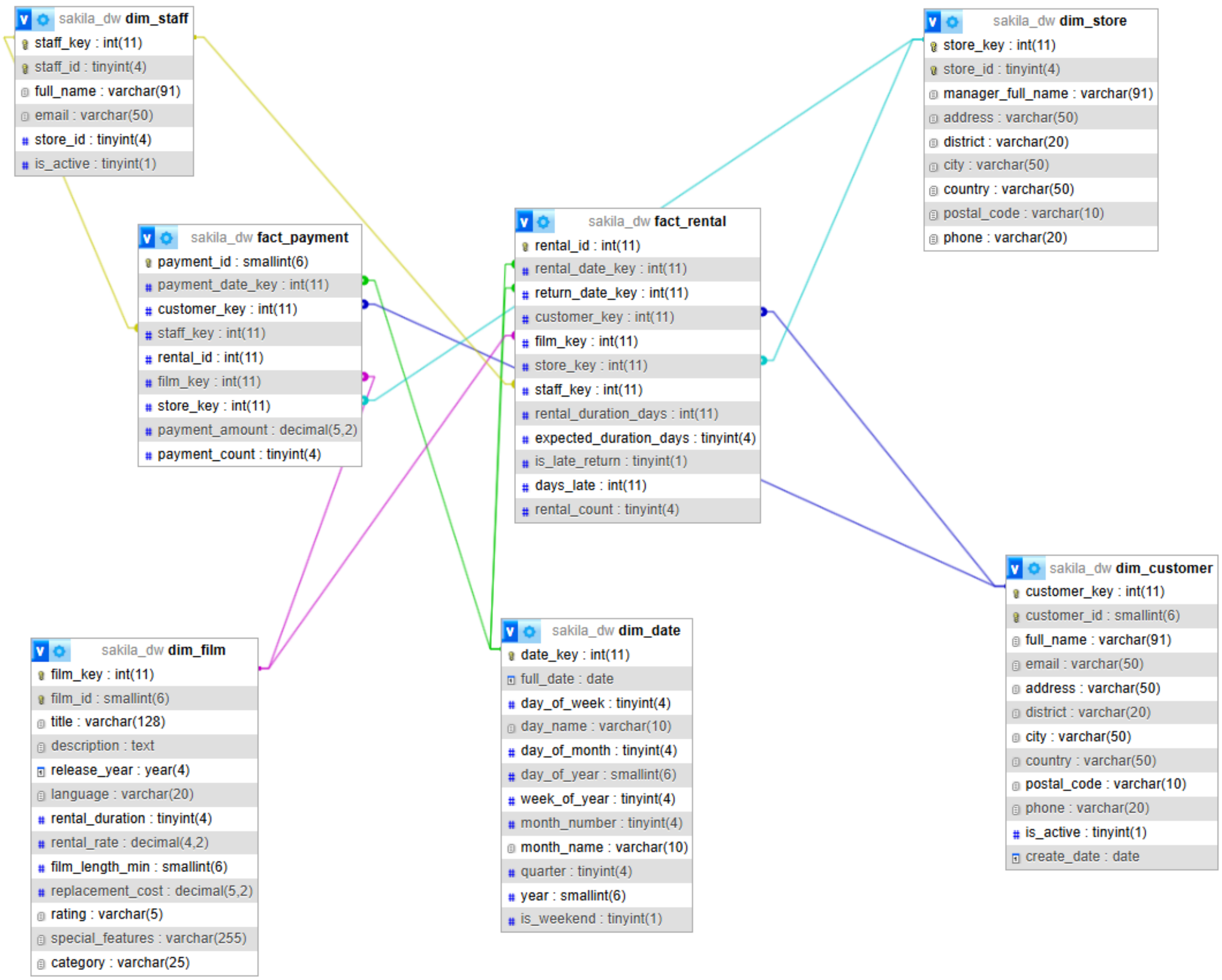
3.3 Dimension Tables

Dimension	Rows Loaded	Key Attributes	OLTP Source Tables
dim_date	1,095 days	date_key, day/month/quarter/year, is_weekend	Generated (no OLTP source)
dim_customer	599 customers	full_name, email, city, country, is_active	customer, address, city, country
dim_film	1,000 films	title, category, language, rental_rate, rating	film, language, film_category, category

dim_store	2 stores	store_id, manager_full_name, city, country	store, staff, address, city, country
dim_staff	2 staff	full_name, email, store_id, is_active	staff

4. Dimensional Model Diagram

The star schema below shows both fact tables sharing all five conformed dimension tables.



5. ETL Design

The ETL pipeline is implemented in Python (movie_rental_dw.ipynb) using the mysql-connector-python library. It runs in 7 sequential steps.

5.1 Extract — Source Tables Used

OLTP Table	Purpose
rental	Primary source for fact_rental. Provides rental_date, return_date, customer_id, staff_id, inventory_id.
payment	Primary source for fact_payment. Provides payment_amount, payment_date, customer_id, staff_id.
inventory	Bridge table linking rental → film and store. Provides film_id and store_id.
film	Core source for dim_film. Provides title, rating, rental_duration, rental_rate, language_id.
film_category	Many-to-many bridge resolved in ETL: one primary category selected per film using ROW_NUMBER().
category	Provides category name (Action, Comedy, etc.) joined via film_category.
language	Provides film language name for dim_film.
customer	Core source for dim_customer. Provides name, email, address_id, active status.
staff	Source for dim_staff and manager name in dim_store.
store	Source for dim_store. Provides manager_staff_id and address_id.
address / city / country	3-table geography chain denormalised into flat city and country columns in dim_customer and dim_store.

5.2 Transform Key Transformations

- Name concatenation: CONCAT(first_name, ' ', last_name) applied to customer, staff, and store manager fields to produce a single full_name column.
- Geography denormalisation: The 4-table chain (customer → address → city → country) collapsed into a single JOIN, producing flat city and country columns in dim_customer and dim_store.
- Date key generation: rental_date, return_date, and payment_date converted to YYYYMMDD integer keys (e.g. 20050624) using Python strftime('%Y%m%d') for efficient joining to dim_date.
- Rental duration: actual rental_duration_days = (return_date − rental_date).days. Expected duration comes from film.rental_duration.
- Late return detection: is_late_return = 1 if actual_days > expected_days. days_late = actual_days − expected_days (negative means early return).
- NULL handling: return_date_key, rental_duration_days, is_late_return, and days_late are set to None (NULL) when return_date is missing meaning the film has not yet been returned.

- Many-to-many resolution: film_category is a many-to-many table. ROW_NUMBER() OVER (PARTITION BY film_id ORDER BY category name) selects one primary category per film to avoid fan-out inflation in fact aggregations.
- SET type conversion: MySQL special_features column is a SET type, received by Python as a set object. Converted to a comma-separated string using ', '.join(sorted(row)) before inserting into dim_film.
- Surrogate key mapping: After loading each dimension, the ETL builds a Python dictionary (natural_key → surrogate_key) to resolve foreign keys when populating fact tables.

5.3 Load — Loading Order and Strategy

Dimension tables are always loaded before fact tables because fact table foreign keys reference dimension surrogate keys that must already exist.

Order	Table	Rows Loaded	Key Strategy
1	dim_date	1,095	Generated in Python — no OLTP source
2	dim_customer	599	AUTO_INCREMENT surrogate key; customer_id retained as natural key
3	dim_film	1,000	AUTO_INCREMENT surrogate key; SET type converted to string
4	dim_store	2	AUTO_INCREMENT surrogate key; manager joined from staff table
5	dim_staff	2	AUTO_INCREMENT surrogate key
6	fact_rental	16,044	Surrogate keys resolved via Python dict lookups; measures calculated
7	fact_payment	16,044	Surrogate keys resolved; film/store context from rental→inventory path

6. Data Quality Results

Seven data quality checks were run automatically at the end of the ETL. Results from the actual pipeline run:

Check	Rule	Result	Status
QC-1	fact_rental row count matches sakila.rental	16,044 = 16,044	**PASS**
QC-2	fact_payment row count matches sakila.payment	16,044 = 16,044	**PASS**
QC-3	No negative rental durations (return before rental date)	0 issues	**PASS**
QC-4	No payment amounts <= 0	24 payments = $0	**WARN**
QC-5	No extreme rentals > 365 days	0 issues	**PASS**
QC-6	No NULL film keys in fact_rental	0 issues	**PASS**
QC-7	No orphan date keys (all dates exist in dim_date)	0 issues	**PASS**

QC-4 Note: The 24 payments with amount = 0$ are present in the original Sakila OLTP data (promotional free rentals). This is a known data characteristic, not an ETL error. The pipeline correctly flags it as a WARN rather than silently ignoring it.

7. Analytical Query Results

All 11 business questions were answered by running movie_rental_dw.ipynb against sakila_dw. Results below are from the actual pipeline run.

BQ-1: Top 10 Most Frequently Rented Films

Title	Category	Total Rentals
BUCKET BROTHERHOOD	Travel	34
ROCKETEER MOTHER	Foreign	33
FORWARD TEMPLE	Games	32
GRIT CLOCKWORK	Games	32
JUGGLER HARDLY	Animation	32
RIDGEMONT SUBMARINE	New	32
SCALAWAG DUCK	Music	32
TIMBERLAND SKY	Classics	31
ZORRO ARK	Comedy	31
GOODFELLAS SALUTE	Sci-Fi	31

Insight: Travel and Games categories dominate the top rentals. BUCKET BROTHERHOOD leads with 34 rentals.

BQ-2: Top 10 Films by Revenue

Title	Category	Total Revenue ($)
TELEGRAPH VOYAGE	Music	231.73
WIFE TURN	Documentary	223.69
ZORRO ARK	Comedy	214.69
GOODFELLAS SALUTE	Sci-Fi	209.69
SATURDAY LAMBS	Sports	204.72
TITANS JERK	Sci-Fi	201.71
TORQUE BOUND	Drama	198.72
HARRY IDAHO	Drama	195.70
INNOCENT USUAL	Foreign	191.74
HUSTLER PARTY	Comedy	190.78

Insight: TELEGRAPH VOYAGE generates the highest revenue ($231.73) despite not being the most rented indicating a higher rental rate.

BQ-3: Film Categories by Popularity and Revenue

Category	Total Rentals	Total Revenue ($)
Sports	1,179	5,314.21
Animation	1,166	4,656.30
Action	1,112	4,375.85
Sci-Fi	1,101	4,756.98
Family	1,096	4,226.07
Drama	1,060	4,587.39
Documentary	1,050	4,217.52
Foreign	1,033	4,270.67
Games	969	4,281.33
Children	945	3,655.55

Insight: Sports is the most popular category with 1,179 rentals and $5,314.21 revenue. Sci-Fi ranks 4th in rentals but 2nd in revenue, suggesting higher pricing.

BQ-4: Store Performance

Store ID	City	Country	Total Rentals	Total Revenue ($)
2	Woodridge	Australia	8,121	33,726.77
1	Lethbridge	Canada	7,923	33,679.79

Insight: Store 2 (Woodridge, Australia) slightly outperforms Store 1 in both rentals and revenue. Performance is very balanced between the two stores.

BQ-5: Top 10 Customers by Rentals and Revenue

Customer	City	Country	Rentals	Revenue ($)
ELEANOR HUNT	Saint-Denis	Réunion	46	216.54
KARL SEAL	Cape Coral	United States	45	221.55
MARCIA DEAN	Tanza	Philippines	42	175.58
CLARA SHAW	Molodetšno	Belarus	42	195.58
TAMMY SANDERS	Changhwa	Taiwan	41	155.59

WESLEY BULL	Ourense	Spain	40	177.60
SUE PETERS	Changzhou	China	40	154.60
RHONDA KENNEDY	Apeldoorn	Netherlands	39	194.61
TIM CARY	Bijapur	India	39	175.61
MARION SNYDER	Santa Barbara	Brazil	39	194.61

BQ-6: Monthly Rental Activity and Revenue

Year	Month	Rentals	Revenue ($)
2005	May	1,156	4,823.44
2005	June	2,311	9,629.89
2005	July	6,709	28,368.91
2005	August	5,686	24,070.14
2006	February	182	514.18

Insight: July 2005 was the peak month with 6,709 rentals and $28,368 revenue. Activity dropped sharply in 2006, indicating the dataset covers only a partial operational period.

BQ-7: Staff Performance

Staff Member	Rentals Processed	Payments Collected ($)
Mike Hillyer	8,040	33,524.62
Jon Stephens	8,004	33,881.94

Insight: Both staff members have almost identical workload. Jon Stephens collected slightly more revenue despite fewer rentals.

BQ-8: Top 10 Cities by Customer Activity

Country	City	Unique Customers	Total Rentals
United States	Aurora	2	50
United Kingdom	London	2	48
Réunion	Saint-Denis	1	46
United States	Cape Coral	1	45
Belarus	Molodetšno	1	42
Philippines	Tanza	1	42
Taiwan	Changhwa	1	41
China	Changzhou	1	40

Spain	Ourense	1	40
Brazil	Santa Barbara	1	39

BQ-9: Average Rental Duration by Category

Category	Avg Actual Days	Avg Allowed Days
Sports	5.20	4.74
Games	5.20	5.01
Comedy	5.15	4.80
Music	5.07	5.26
Sci-Fi	5.06	4.78
Documentary	5.06	4.65
Family	5.05	5.16
Horror	5.05	4.93
Foreign	5.04	5.11
Action	5.03	4.94

Insight: Sports and Games customers keep films longest (5.20 days avg) — both exceeding their allowed rental duration, meaning late returns are common in these categories.

BQ-10: Films Returned Late Most Often

Title	Category	Rentals	Late Returns	Late %
RIDGEMONT SUBMARINE	New	31	26	83.9%
TELEGRAPH VOYAGE	Music	27	24	88.9%
BUTTERFLY CHOCOLAT	New	30	24	80.0%
GRIT CLOCKWORK	Games	32	23	71.9%
ROCKETEER MOTHER	Foreign	33	22	66.7%
TIMBERLAND SKY	Classics	31	22	71.0%
CHANCE RESURRECTION	Sports	27	22	81.5%
ENGLISH BULWORTH	Sci-Fi	30	21	70.0%
WIFE TURN	Documentary	31	21	67.7%
EXPENDABLE STALLION	Documentary	28	21	75.0%

Insight: TELEGRAPH VOYAGE has the highest late return rate at 88.9%. These films may need a longer allowed rental duration or higher late fees.

BQ-11: Quarterly Revenue Trend

Year	Quarter	Revenue ($)
2005	Q2	14,453.33
2005	Q3	52,439.05
2006	Q1	514.18

Insight: Q3 2005 generated by far the highest revenue at $52,439. Q2 2005 and Q1 2006 show the business ramp-up and wind-down periods in the dataset.

8. Performance Benchmark

To evaluate the effectiveness of the dimensional model, several analytical queries were executed on both the original OLTP Sakila database and the proposed data warehouse.

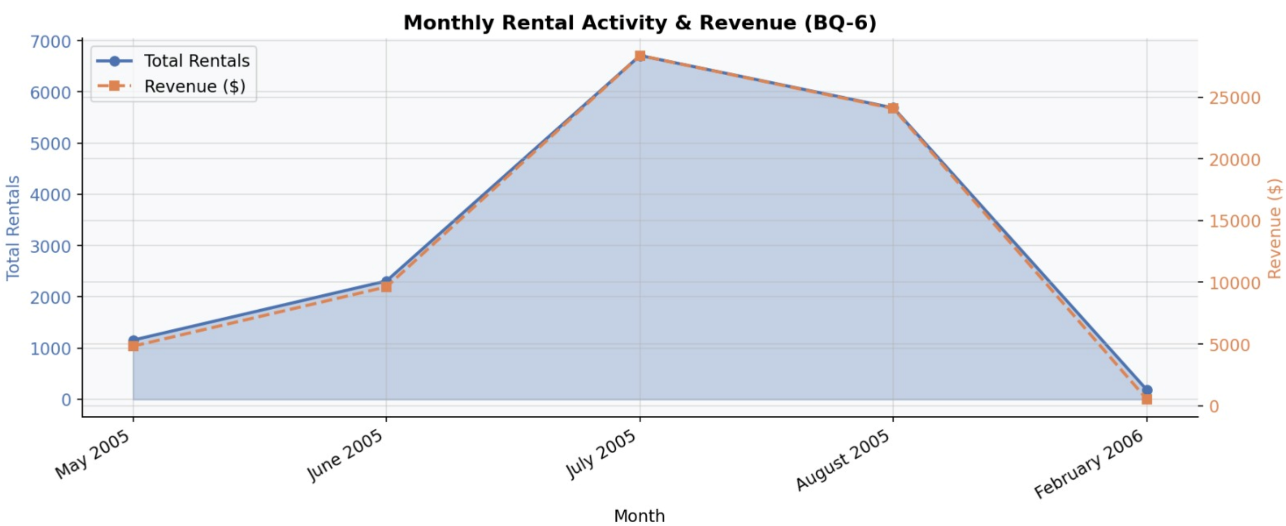
Execution times were measured multiple times and the median execution time was recorded.

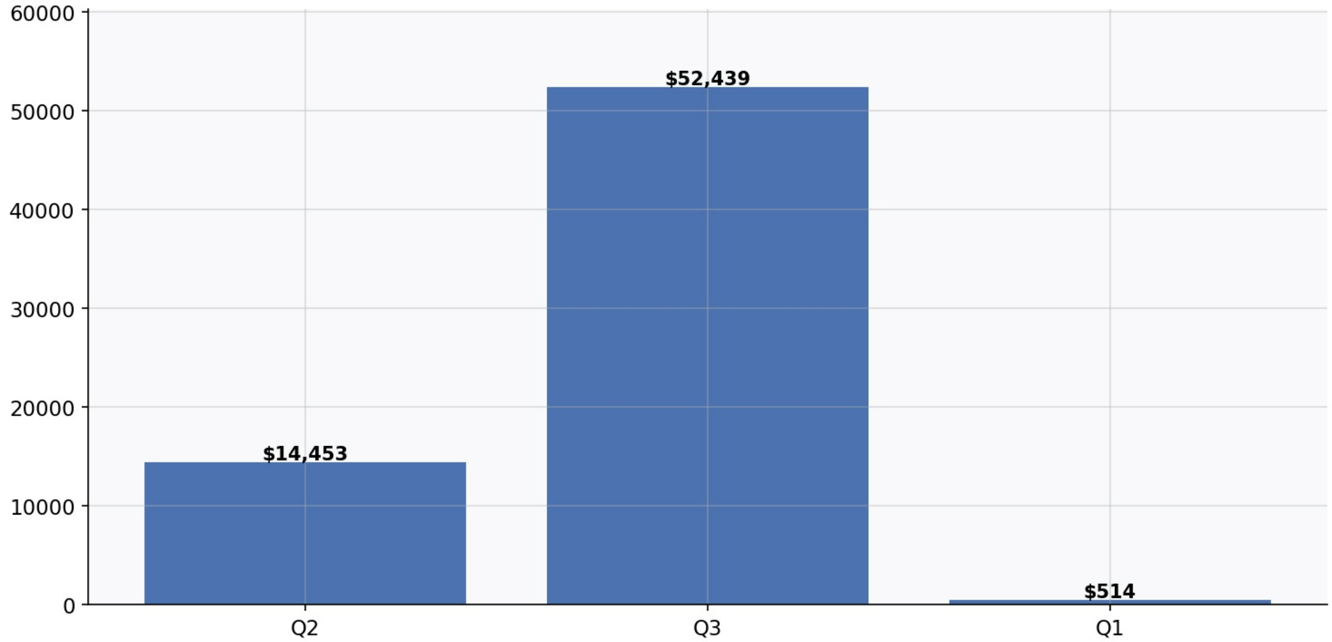
Query	OLTP (ms)	DW (ms)	Speedup
BQ-1 Top Rented Films	29.07	32.80	0.9x
BQ-2 Film Revenue	64.33	43.08	1.5x
BQ-3 Category Popularity	61.46	67.04	0.9x
BQ-6 Monthly Trends	48.60	72.75	0.7x
BQ-10 Late Returns	54.44	31.32	1.7x

The benchmark demonstrates that the dimensional warehouse performs particularly well for aggregation-heavy analytical workloads such as revenue analysis and late-return detection.

Some queries remain similar in performance because the Sakila dataset is relatively small. In real-world environments with millions of records, dimensional models typically provide larger performance improvements due to denormalization, indexing, and simplified analytical joins.

9. Visualizations





10. Conclusion

This project successfully implemented a complete data warehouse pipeline for the Sakila movie rental OLTP database:

- The Sakila database (16 OLTP tables) was imported into XAMPP MySQL and verified.
- A star schema (sakila_dw) was designed with 2 fact tables and 5 conformed dimension tables.
- A Python ETL script loaded 16,044 rental facts, 16,044 payment facts, 1,000 film dimensions, 599 customer dimensions, and a 1,095-day date dimension.
- Seven data quality checks confirmed 6 PASS results. One WARN (24 zero-amount payments) was identified as a known Sakila data characteristic, not an ETL defect.
- All 11 business questions were answered with real query results, providing actionable insights on film popularity, revenue, customer behaviour, store performance, staff output, and time trends.

The star schema design enables business managers to run fast analytical queries without complex joins, supports conformed dimensions shared across both fact tables, and provides a foundation that can be extended with additional dimensions or fact tables as the business grows.